

Q Search & insert position

T.C $\Rightarrow O(\log n)$

S.C $\Rightarrow O(1)$

→ Code

```
int start = 0;
```

```
int end = arr.length - 1;
```

```
while (start <= end) {
```

```
    int mid = start + (end - start) / 2;
```

```
    if (target == arr[mid]) return mid;
```

```
    else if (arr[mid] < target) start = mid + 1;
```

```
    else end = mid - 1;
```

```
} else start;
```

→ if target is not found this is the idx where it need to be inserted.

→ if target found return idx;